

FANGTONG ZHOU

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EDUCATION

- Virginia Tech**, Blacksburg, VA, USA 2024.8 - present
- Ph.D. in Electrical Engineering
 - GPA: 4.0/4.0
 - Advisor: Prof. Tom Hou
- ShanghaiTech University**, Shanghai, China 2021.9 - 2024.6
- M.S. in Information and Communication Engineering
 - GPA: 3.78/4.0
 - Advisor: Prof. Yang Yang, Prof. Yong Zhou
- South China University of Technology**, Guangzhou, China 2017.9 - 2021.6
- B.Eng. in Information Engineering
 - GPA: 3.75/4.0 (89.00/100)
 - Ranking: 8/233

RESEARCH EXPERIENCE

- **AirComp-assisted Hierarchical Personalized Federated Learning** 2022.6 - 2024.6
 - Proposed an AirComp-assisted hierarchical personalized FL that simultaneously learns global and personalized models while mitigating interference through cloud/edge beamforming design, significantly improving convergence and accuracy in heterogeneous wireless networks
- **Edge Interval Control in Hierarchical Federated Learning** 2023.3 - 2023.6
 - Developed an AirComp-assisted hierarchical FL that jointly optimizes edge aggregation intervals and device transceiver design via relaxation-rounding and Lyapunov-based algorithms, achieving faster convergence and higher accuracy under wireless communication constraints
- **Decentralized Satellite Federated Learning** 2023.6 - present
 - Introduced a multi-orbit decentralized SFL framework that leverages intra- and inter-orbit ISLs for model aggregation without ground stations, analyzes convergence to guide local iteration settings, and develops a JRARS algorithm for joint routing, bandwidth, and power optimization, achieving faster convergence and lower energy consumption in LEO constellations
- **WOS** 2025.1 - 2025.6
 - Proposed Wait-for-Optimal-Set (WOS), a semi-asynchronous federated learning scheme that adaptively selects clients based on computation latency and model-version gap while employing dynamic resource block allocation, achieving faster convergence and higher accuracy than existing methods in dynamic wireless networks
- **FedHusky** 2025.6 - 2025.9
 - Proposed FedHusky, a hybrid FL framework that employs a calendar-based client scheduling, optimization-driven group formation, and dynamic birth-death processes to maximize client utilization, significantly improving convergence speed and training efficiency under small and heterogeneous datasets

PUBLICATIONS

- [1] **F. Zhou**, Z. Wang, X. Luo, and Y. Zhou, **"Over-the-air computation assisted hierarchical personalized federated learning,"** in Proc. IEEE Int. Conf. on Commun. (ICC), Rome, Italy, May 2023.
- [2] **F. Zhou**, X. Chen, H. Shan, and Y. Zhou, **"Adaptive Transceiver Design for Wireless Hierarchical Federated Learning,"** in Proc. IEEE 98th Vehicular Technology Conference (VTC2023-Fall), Hong Kong, China, Oct. 2023.
- [3] L. Wu, G. Gao, J. Yu, **F. Zhou**, Y. Yang, and T. Wang, **"Pdd: Partitioning dag-topology dnns for streaming tasks,"** IEEE Internet of Things Journal, vol. 11, no. 6, pp. 9258–9266, Mar. 2024.
- [4] **F. Zhou**, Z. Wang, H. Shan, L. Wu, and Y. Zhou, **"Over-the-Air Hierarchical Personalized Federated Learning,"** IEEE Transactions on Vehicular Technology, vol. 74, no. 3, pp. 5006–5021, Mar. 2025.
- [5] **F. Zhou**, Z. Wang, Y. Shi, and Y. Zhou, **"Decentralized Satellite Federated Learning via Intra-and Inter-Orbit Communications"** in Proc. IEEE Int. Conf. on Commun. Workshops (ICC Wcshps), Denver, CO, USA, June 2024.
- [6] **F. Zhou**, Y. Shi, Y. Wu, S. Archarya, L. DaSilva, S. Kompella, W. Lou, and Y. T. Hou **"WOS: An Optimized Scheduling Scheme for Federated Learning in Dynamic Wireless Networks"** in Proc. IEEE Military Communications Conf. (MILCOM), Los Angeles, CA, USA, Oct. 2025.

TEACHING ASSISTANT

ShanghaiTech University

CS 287: Network Intelligence, Fall 2022

Virginia Tech

ECE 2714: Signals and Systems, Fall 2024 & Spring 2025

SKILLS

- **Programming**
Python, Matlab
- **Languages**
 - Chinese (Native)
 - English (IELTS: 7.5; -Reading 7.5; -Listening 8.5; -Speaking 8.0; -Writing: 6.5)